animated is an important one when considering immediate-mode graphics. You want to create smooth animations, though. How do you do that?

11.5. Creating animations

The time-honored way to animate in a browser is to use `setInterval` or `setTimeout` to repeatedly change some property of a DOM object over time. Suppose you have a reference named `bar` to a DOM object that you want to animate:

```
seconds = (n) -> n*1000
setInterval bar.height, seconds 1
```

In a retained-mode API like the DOM, you directly manipulate the property on the object. How the object is rendered is taken care of by the retained-mode API. This is in contrast to an immediate-mode API like `canvas` where you do the rendering yourself.

This section explores animations in retained and immediate mode. It also explores the question of how to keep animations smooth in the single-threaded event-loop environment of a web browser.

11.5.1. Retained mode

To animate DOM objects in retained mode you repeatedly change a `style` property. Suppose you want to animate the bar on a bar chart so that instead of changing from a height of 20 to a height of 40 instantaneously (as with listing 11.4), it animates the change smoothly over one second.

Tweens

The creation of animation between two states is called a *tween*.

A single bar in your bar chart is drawn as a single DOM element, and the height is set as a `style` property: